

UPDATE 4:
DEMO, EXPERIMENTS
& ANALYSIS

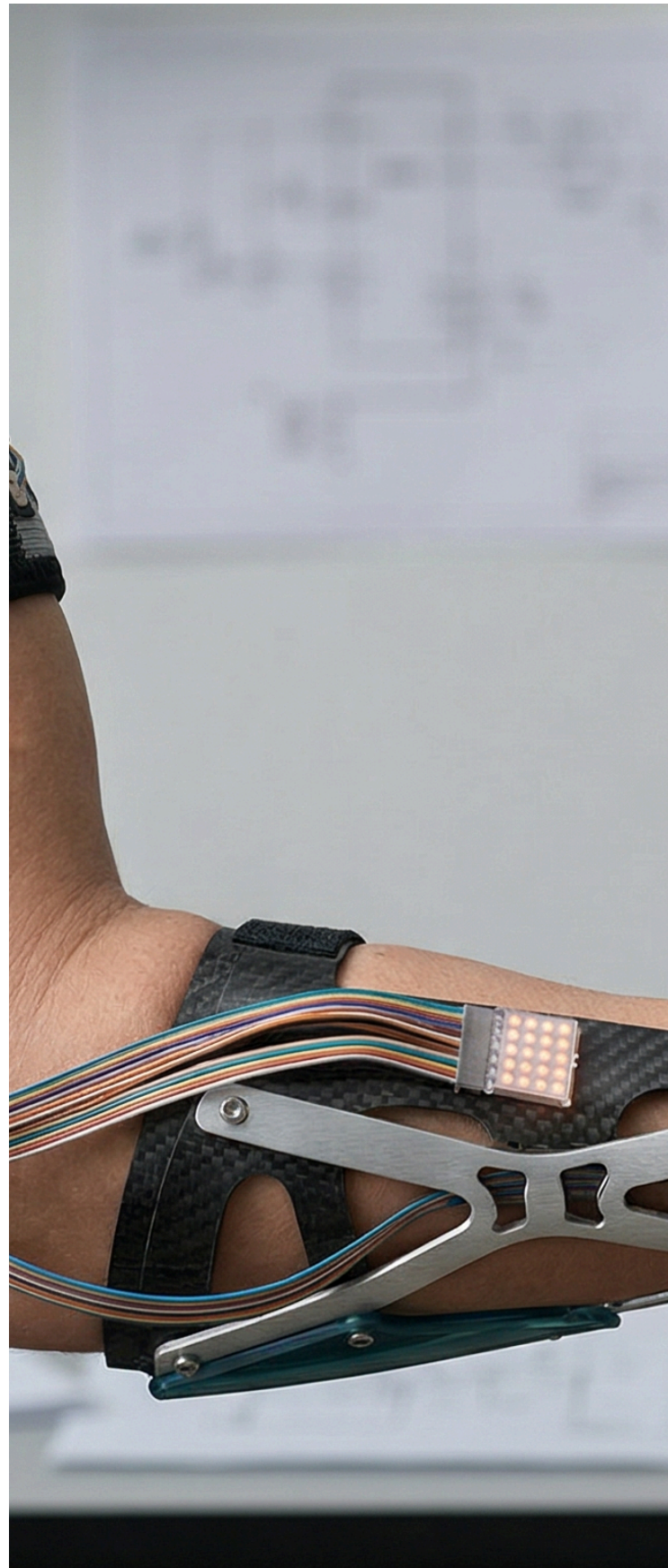
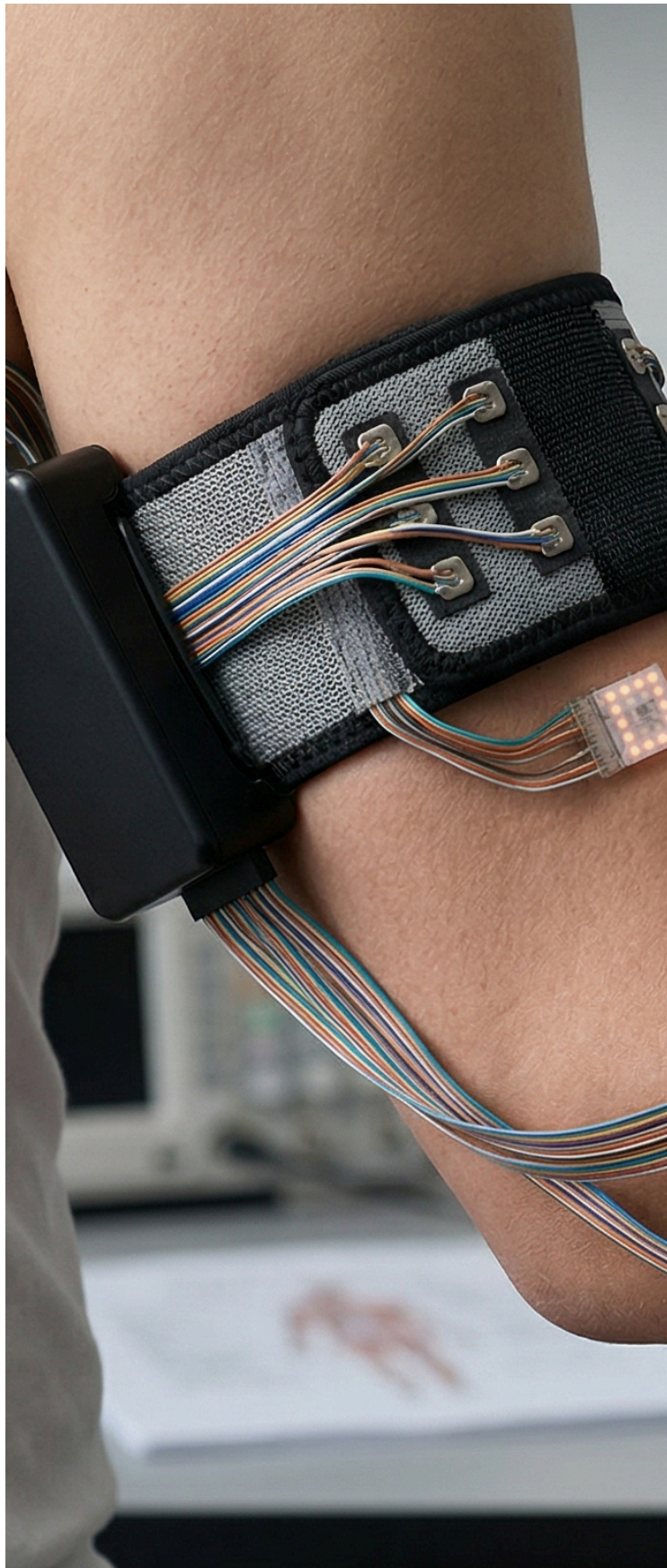
MYOHAP

**Soft Robots for Rehabilitation and
Wearable Exosuits**

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RECAP

Keywords

Prosthesis

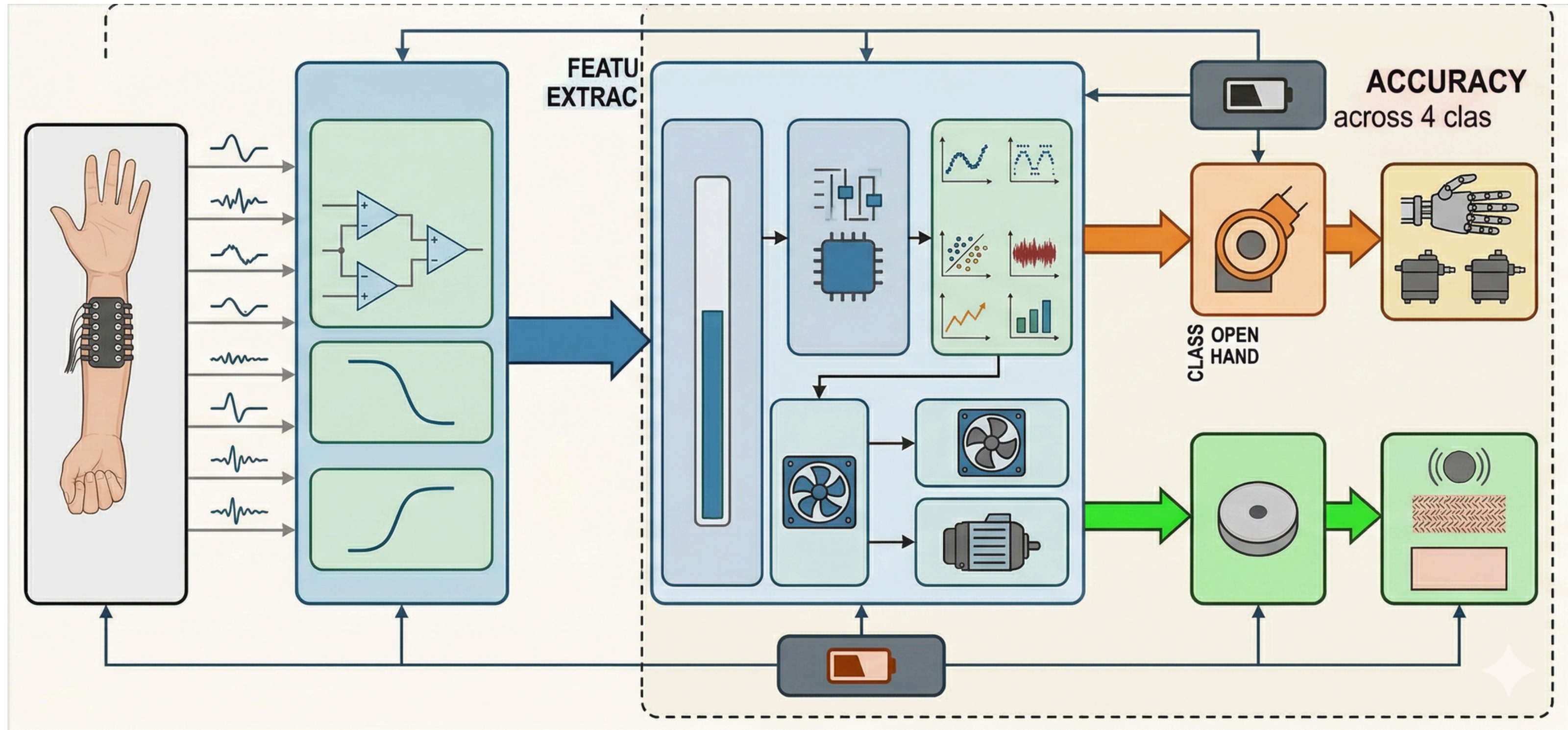
EMG

Intent Detection

Classification of Actions

Haptic Feedback

HIGH LEVEL BREAKDOWN



SYSTEM BREAKDOWN

01. Data Collection

EMG sensors pick up motor neuron activity

02. Filtering

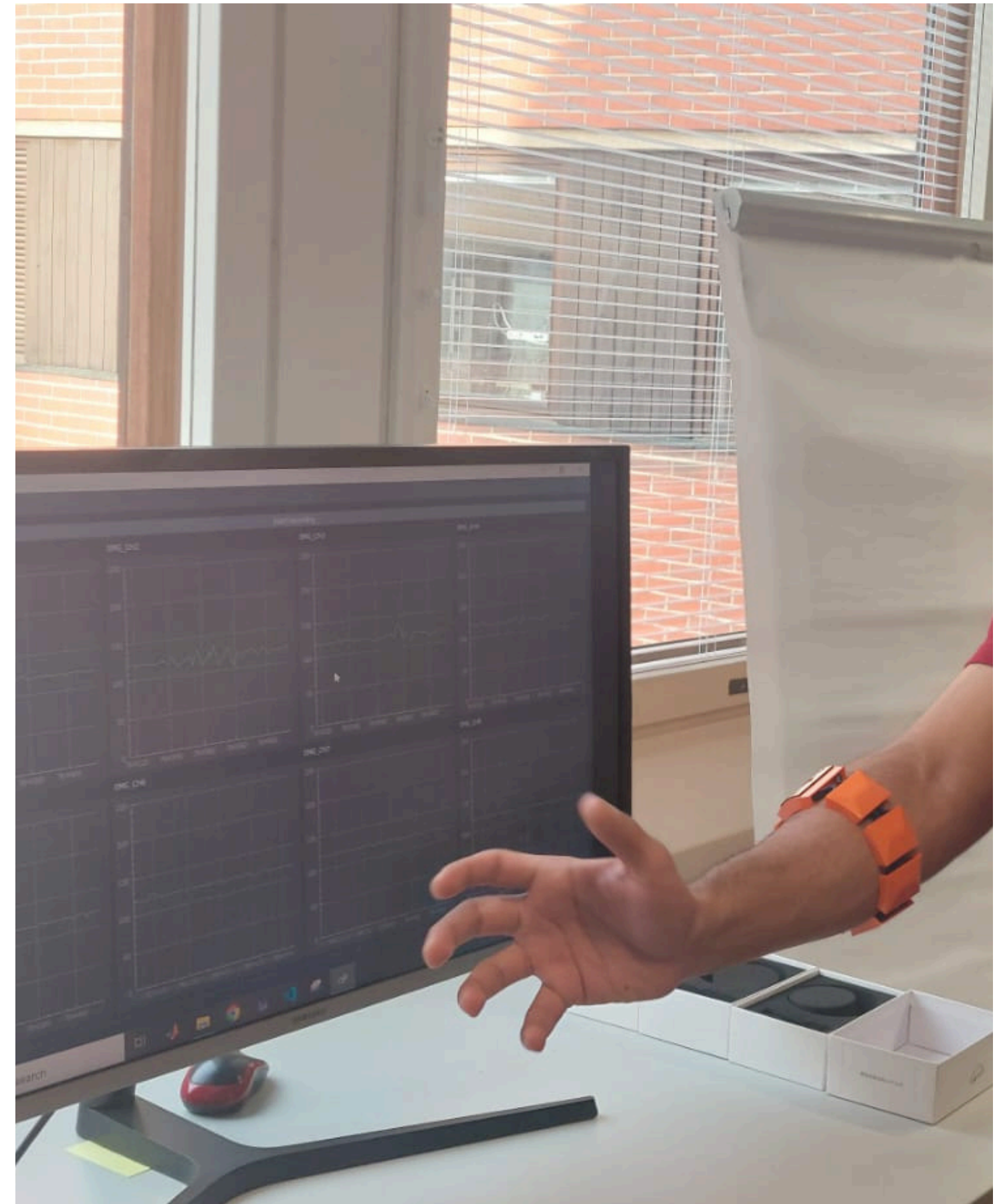
Picked up data is filtered using various filtering techniques

03. Processing

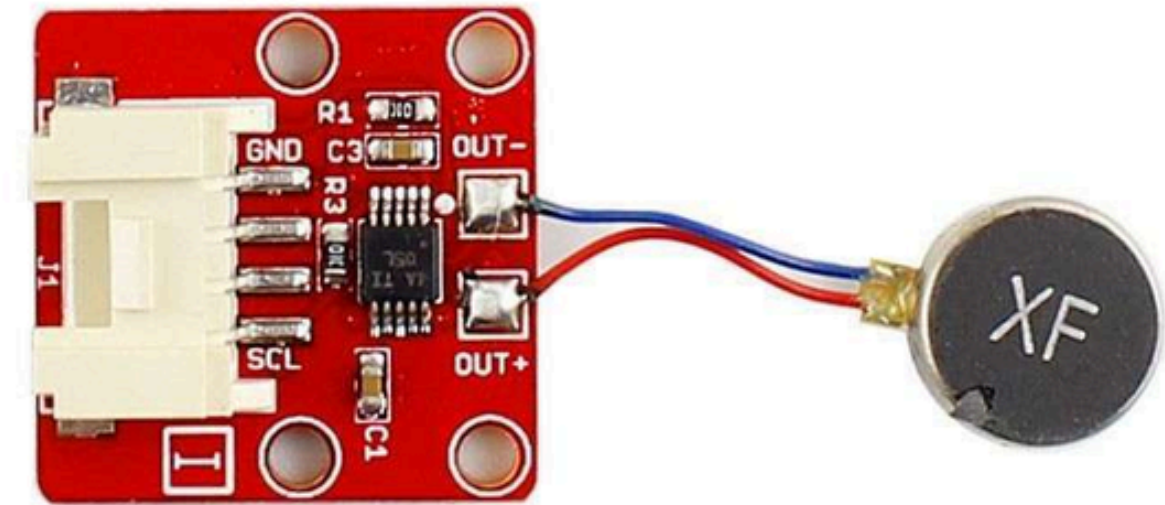
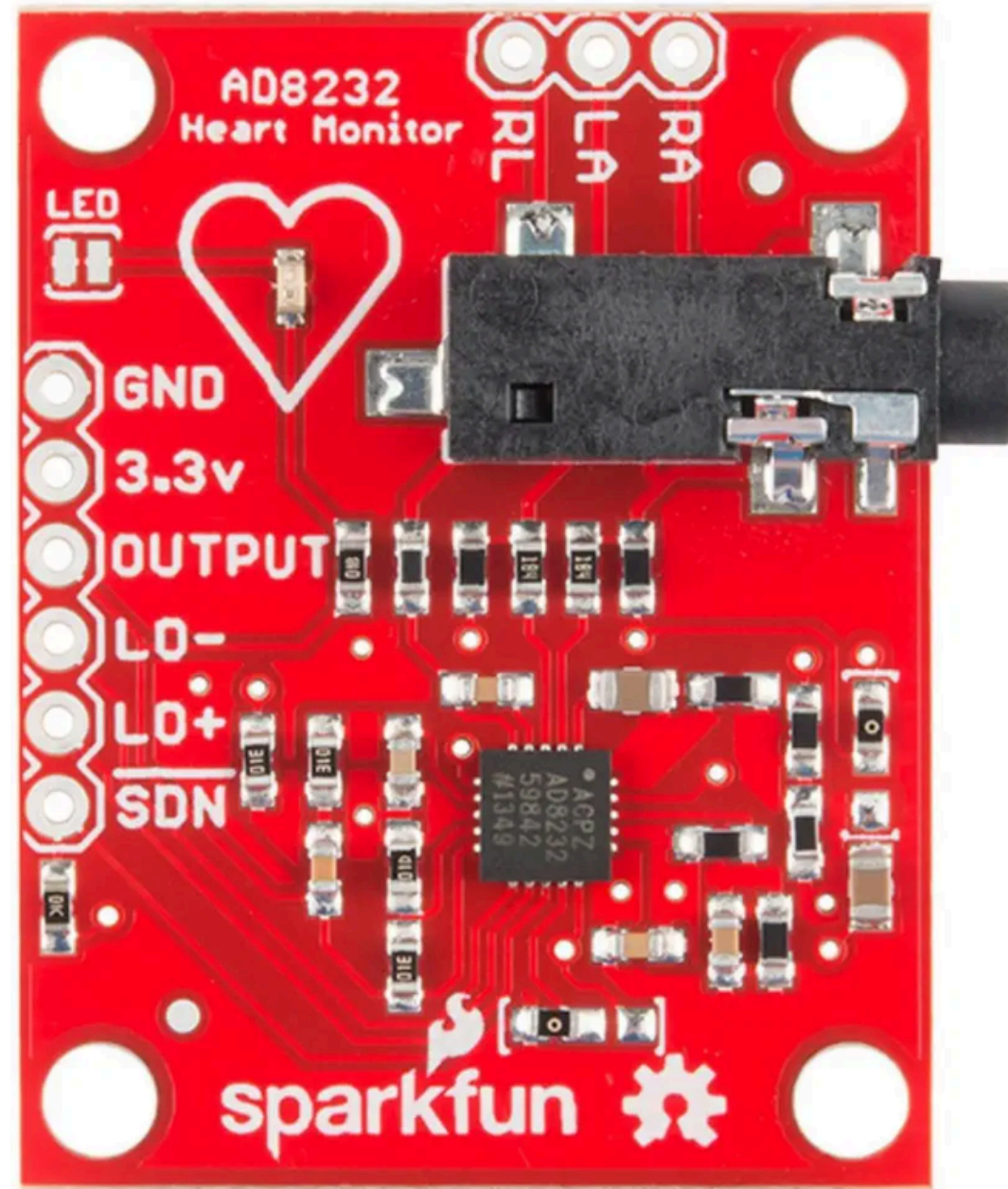
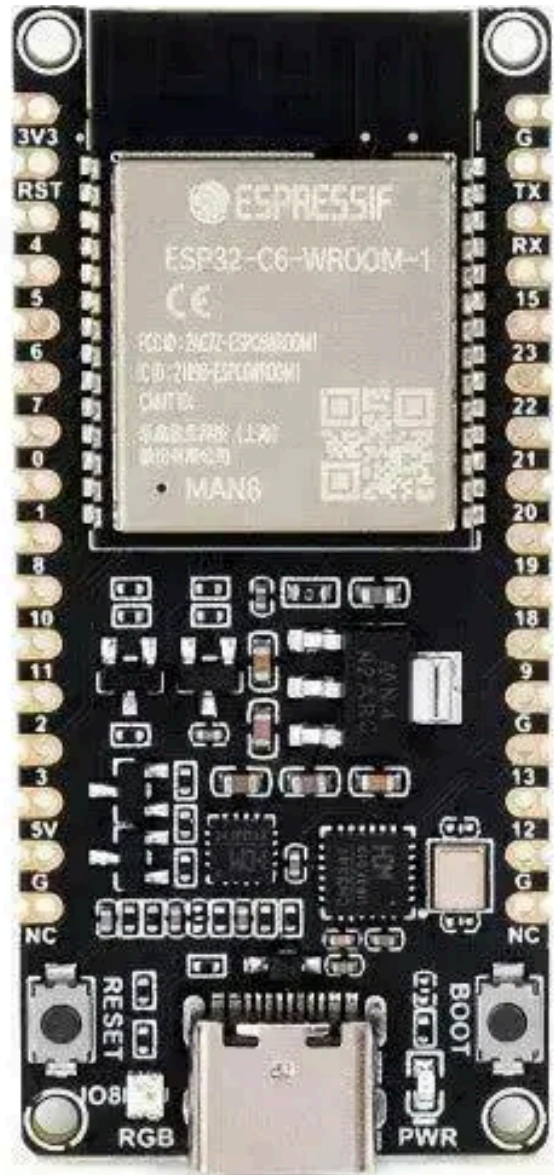
Classification of the EMG signals helps determine what is the action and the intensity of the muscle contraction

04. Encoding

Actuation and Haptic controls are encoded and transmitted through drivers to control the motors



CHOICE OF HARDWARE

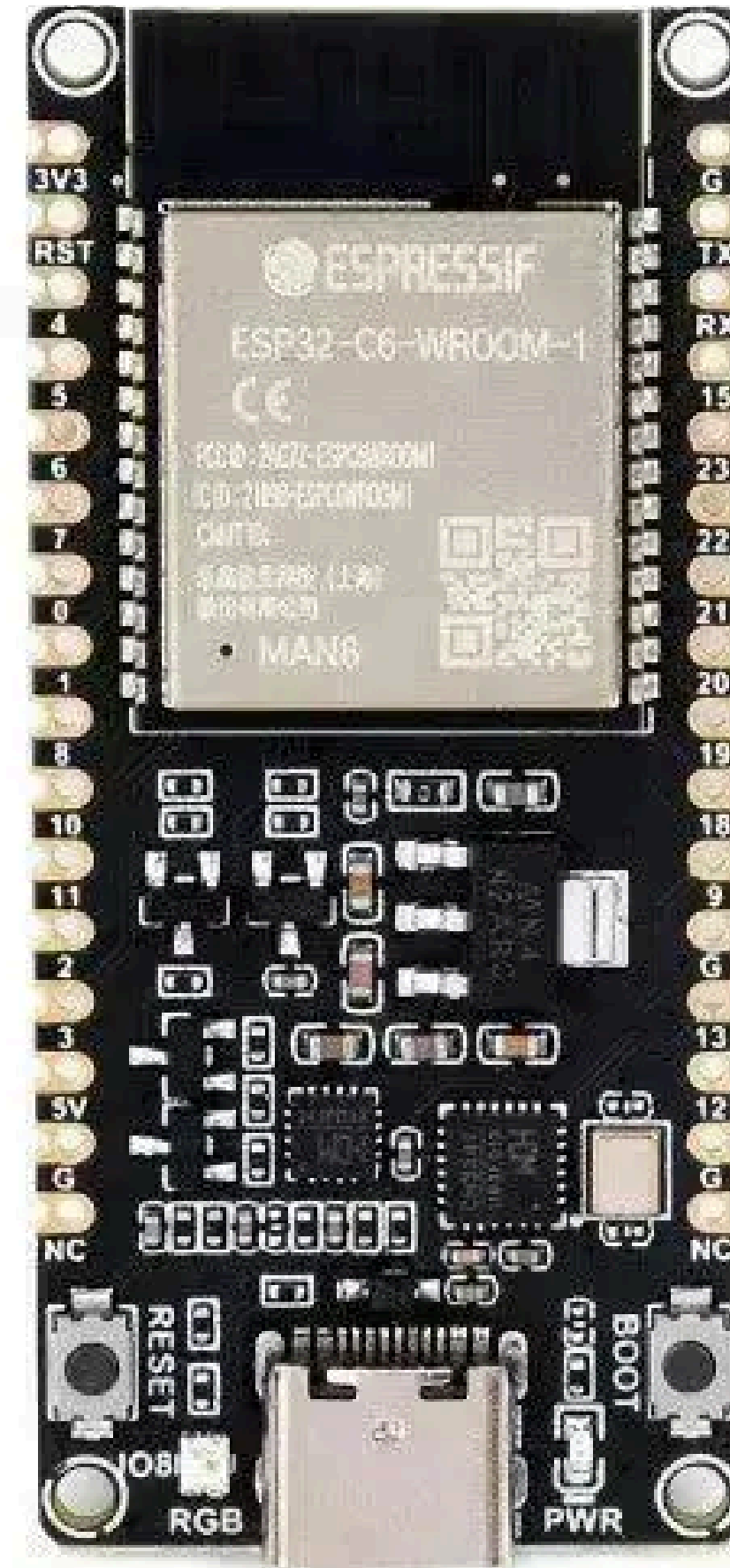


CHOICE OF HARDWARE

ESP32-Microcontroller

The “Brain” of the device. Main uses:

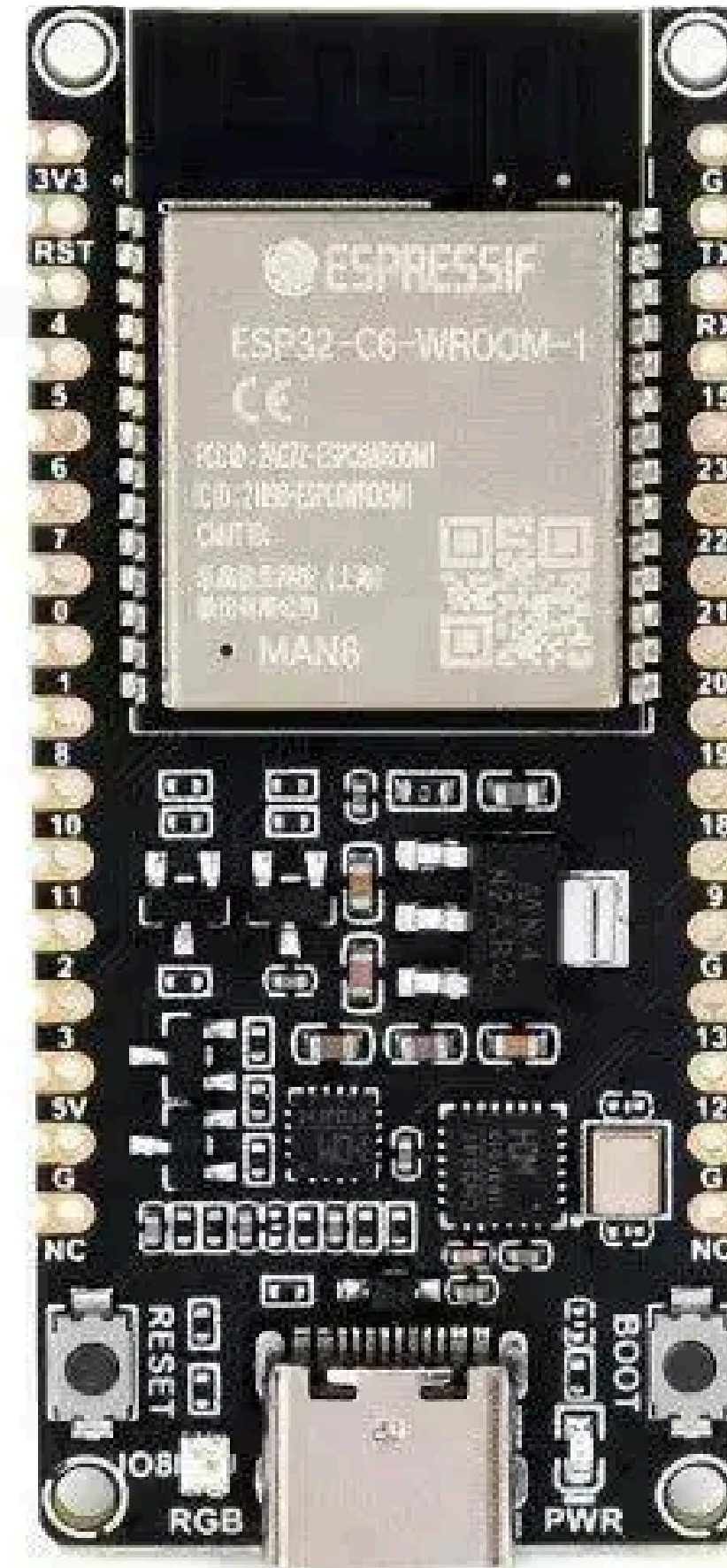
- Central Integration Hub
- High-Speed Data Acquisition (1000Hz)
- Real-Time Signal Processing
- Running Machine Learning (SVM, LDA)
- Managing Closed Loop feedback



CHOICE OF HARDWARE

ESP32-Microcontroller

Parameter	Value
Operating voltage	3.3V
Input voltage	3.3V~5.5V
Support maximum discharge current	600mA@3.3V LDO
Support maximum charge current	500mA
Support USB charging	Yes
Processor	Tensilica LX6 dual core processor
Frequency	240MHz
SRAM	520KB
Flash	16MB

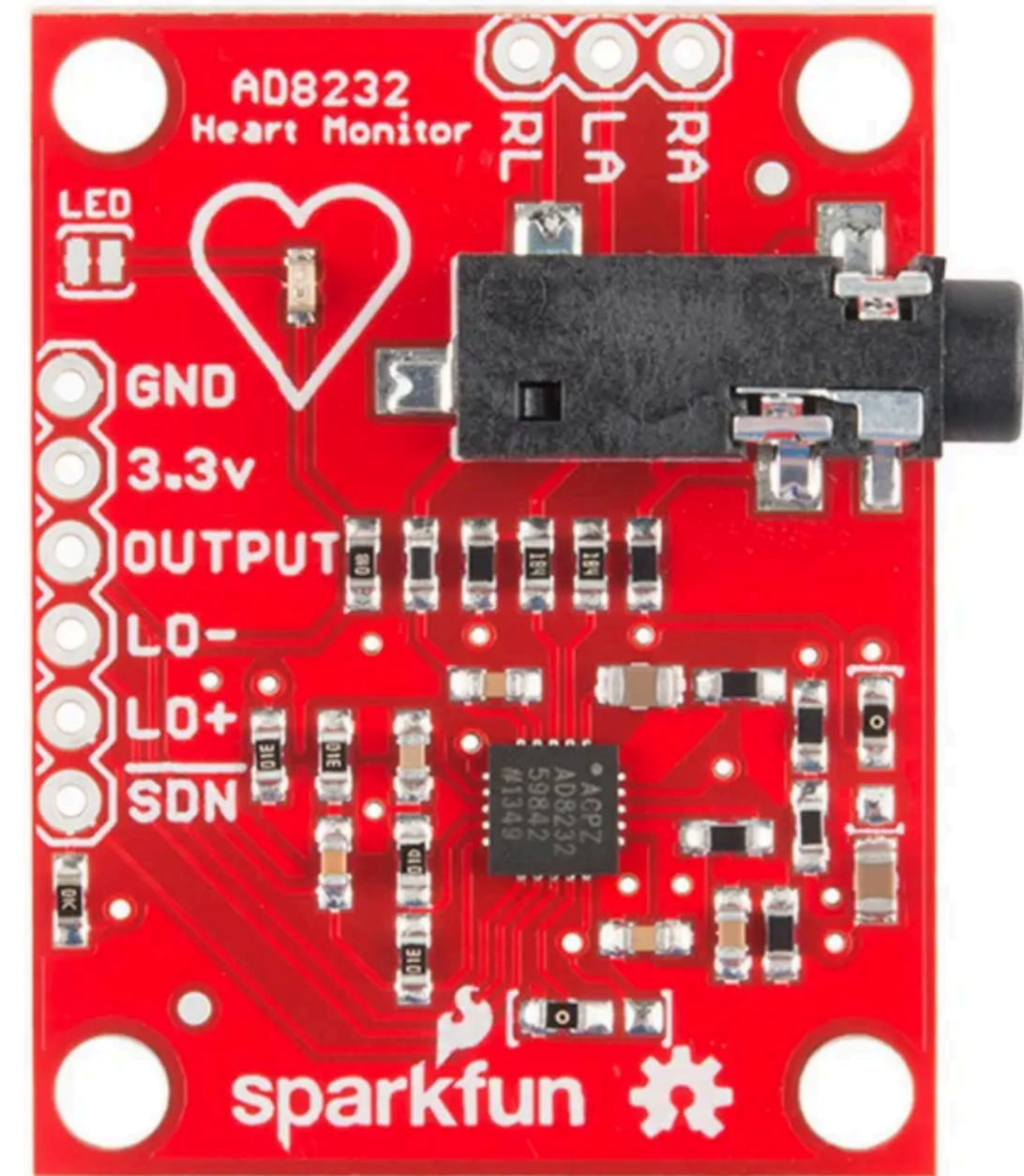


CHOICE OF HARDWARE

AD8232 - ECG Driver

Forms the “Sensory Organ” of the device: Main uses:

- Analog Front End
- High-Gain Amplification
- H/w level smoothening (spiky signal)
- Noise Cancelling (Electrical Interference)
- Safe Voltage Shifting (-ve voltages normalised and rectified)

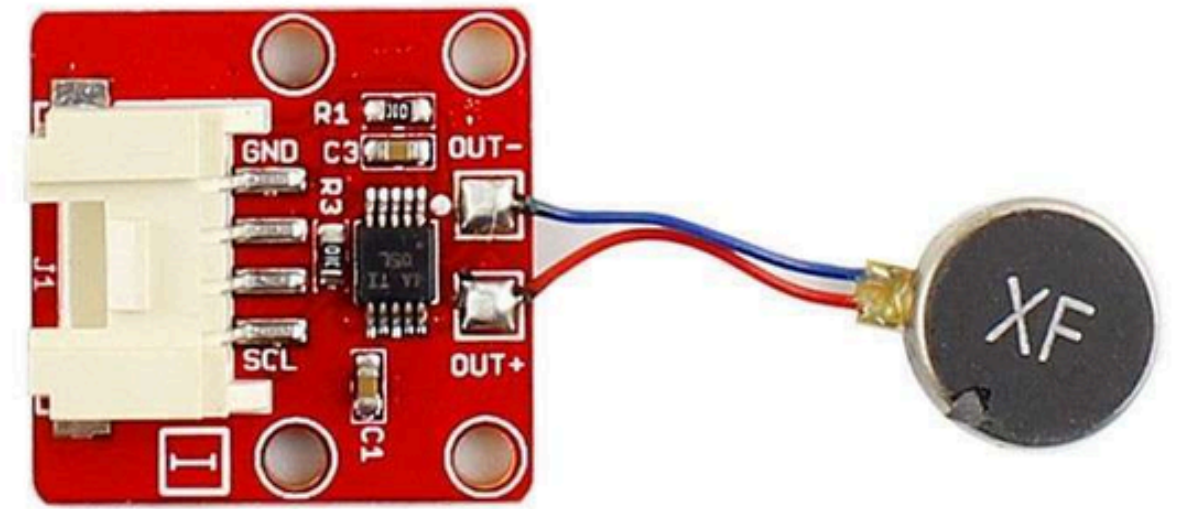


CHOICE OF HARDWARE

CRT00349H - Haptic Device Driver

This is the output other than the actuation. Functions:

- Translation Bridge (Digital to Analog)
- Offloading the (Human) Brain
- Closing the sensory loop



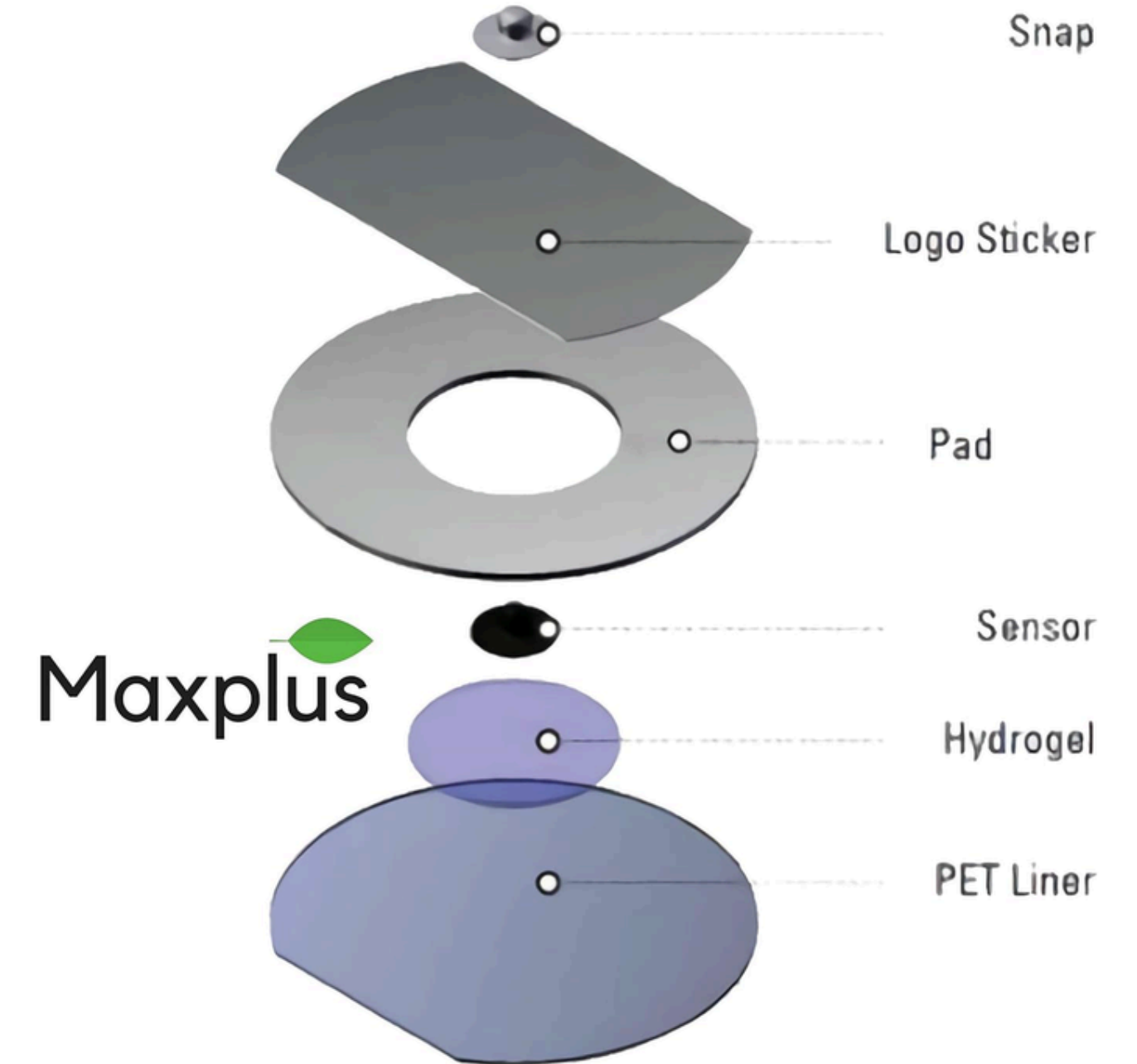
ELECTRODE-EXPLAINED

The **Snap** connector is where the cable connects to AD8232

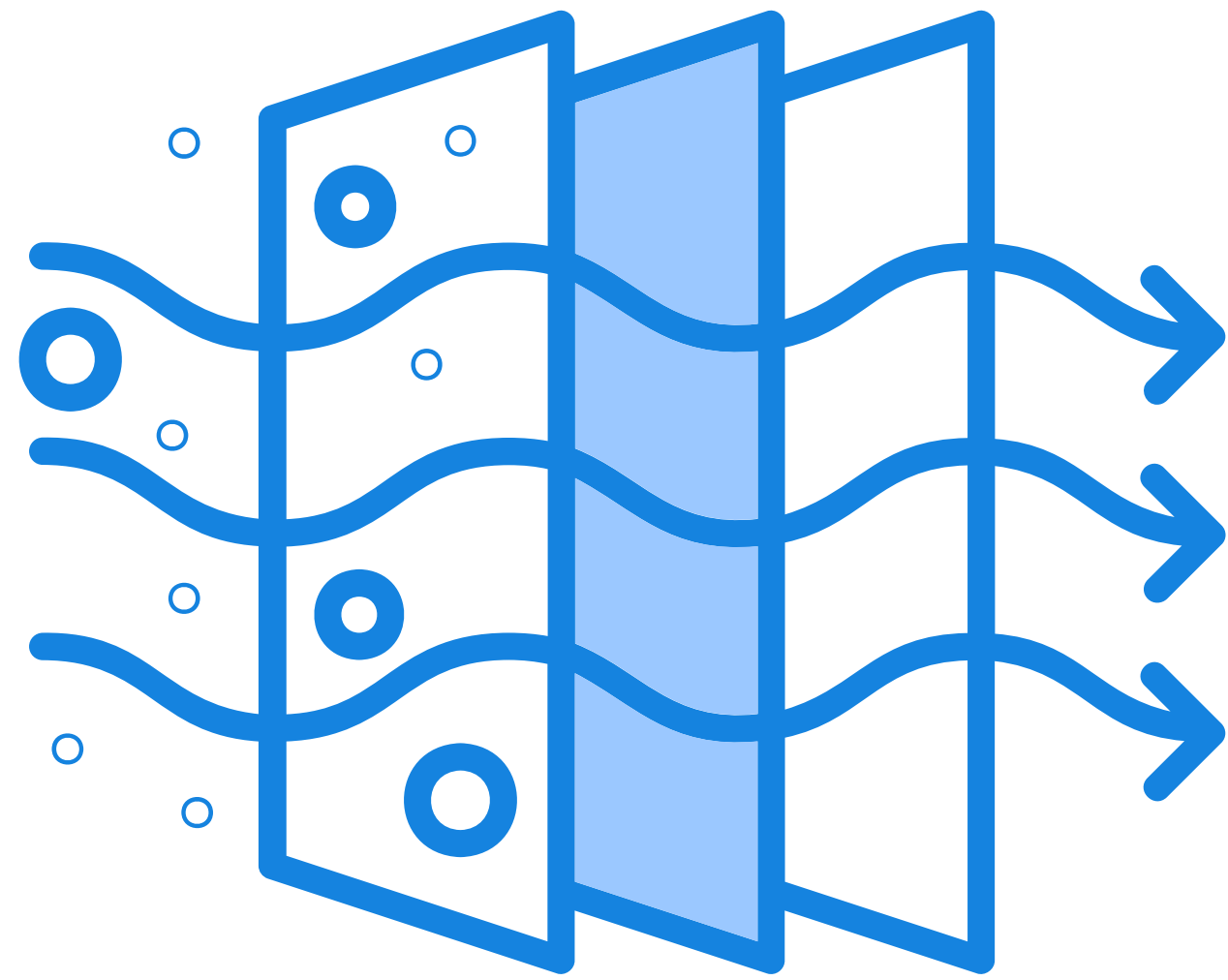
The **pad or Adhesive backing** acts as the anchor for the electrode

The **sensor element**, made from Silver or Silver Chloride, is the main electrode

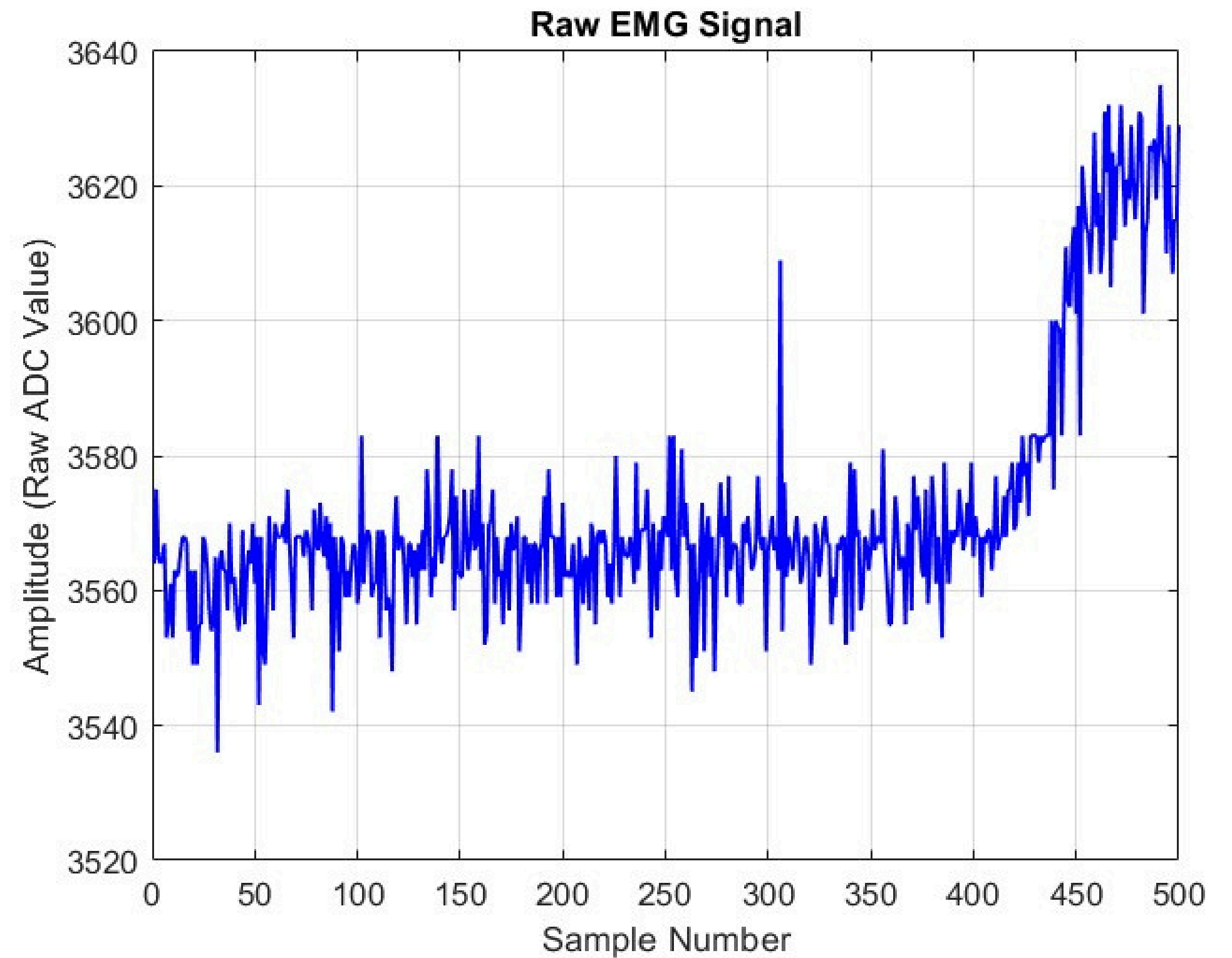
Hydrogel improves conductivity



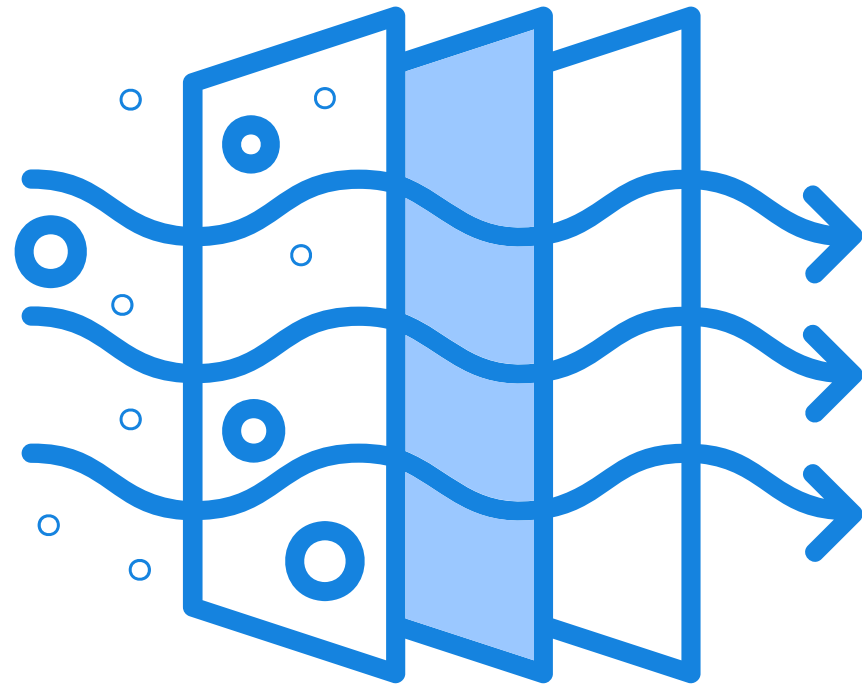
LET'S TALK FILTERING



Signal Noise?



LET'S TALK FILTERING



Body Noise

Other Muscles
Heartbeat

Electrical Interferences

Electrical Outlets
Fluorescent Lights
Chargers
Smart appliances

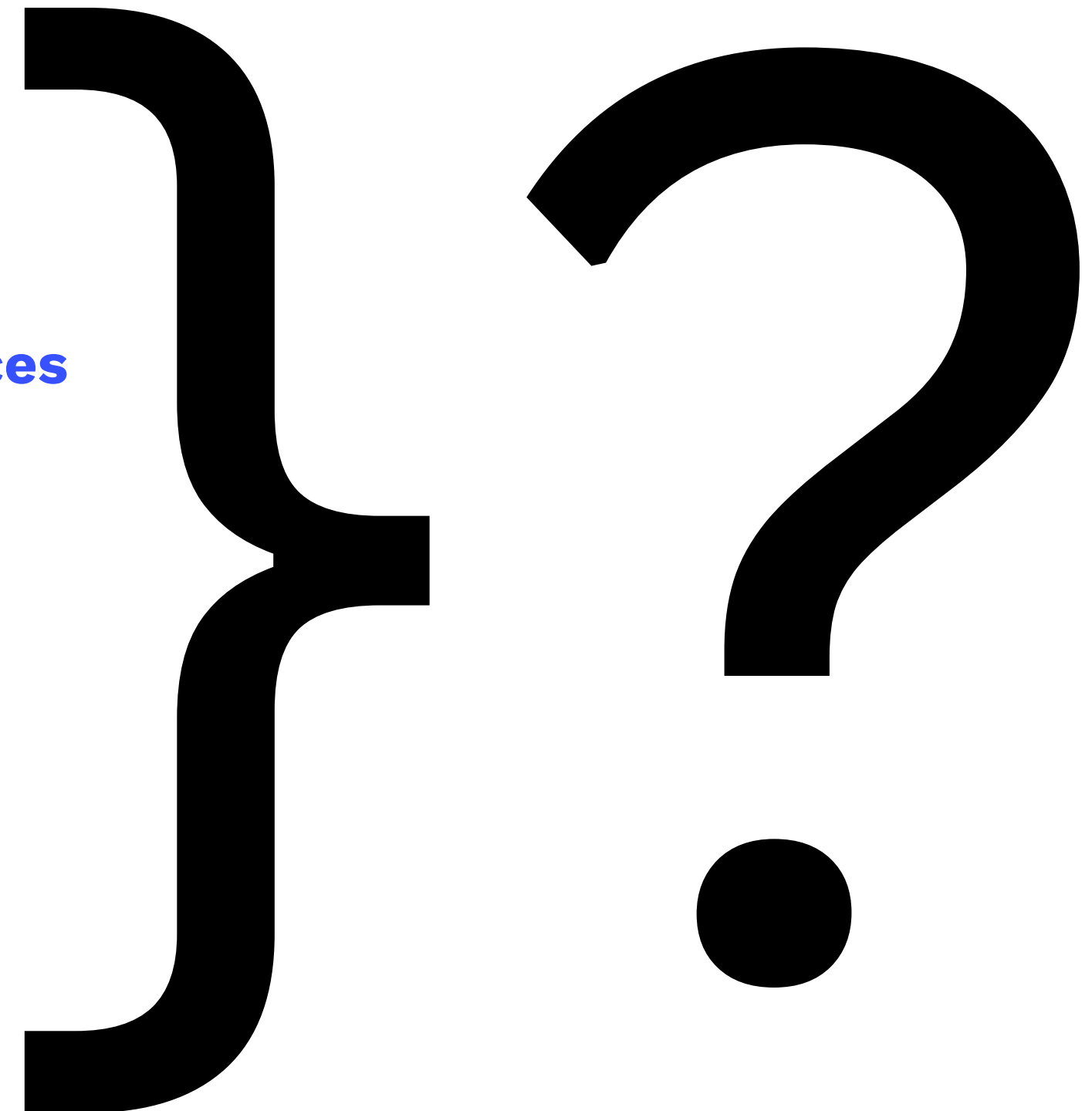
RESOURCES AND CONSTRAINTS

Computation

Very accurate devices need higher computation and more resources like memory and larger buffers

Timing

Real-Time operations mean decisions are made now! Time can't be spent thinking.



FILTERING

Window Size

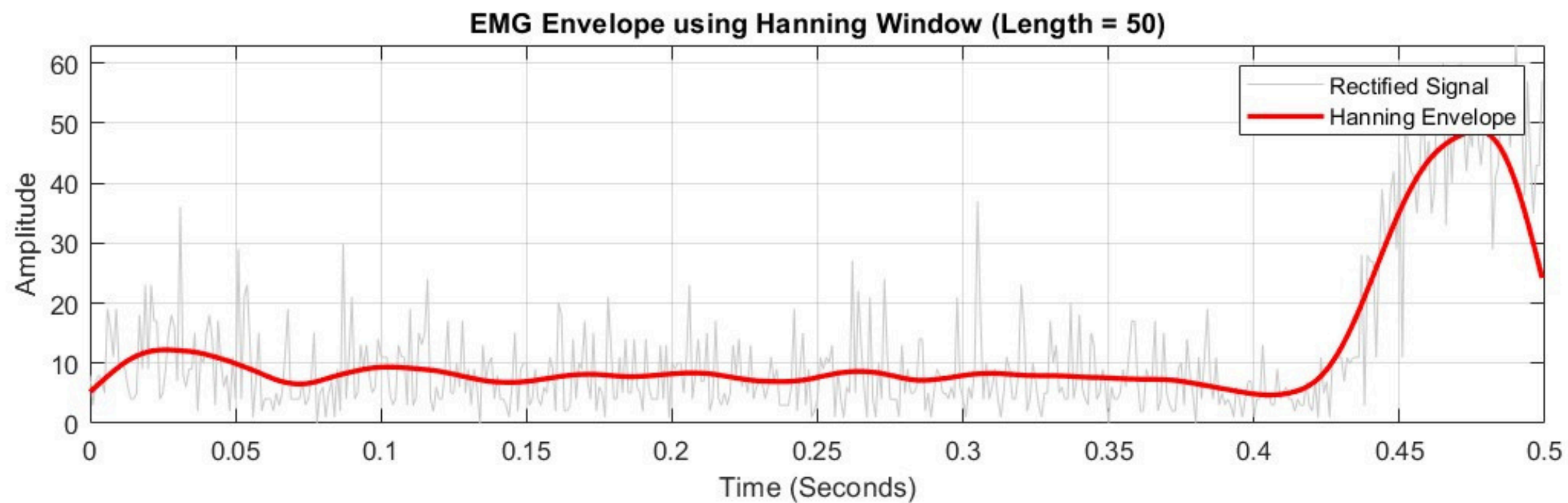
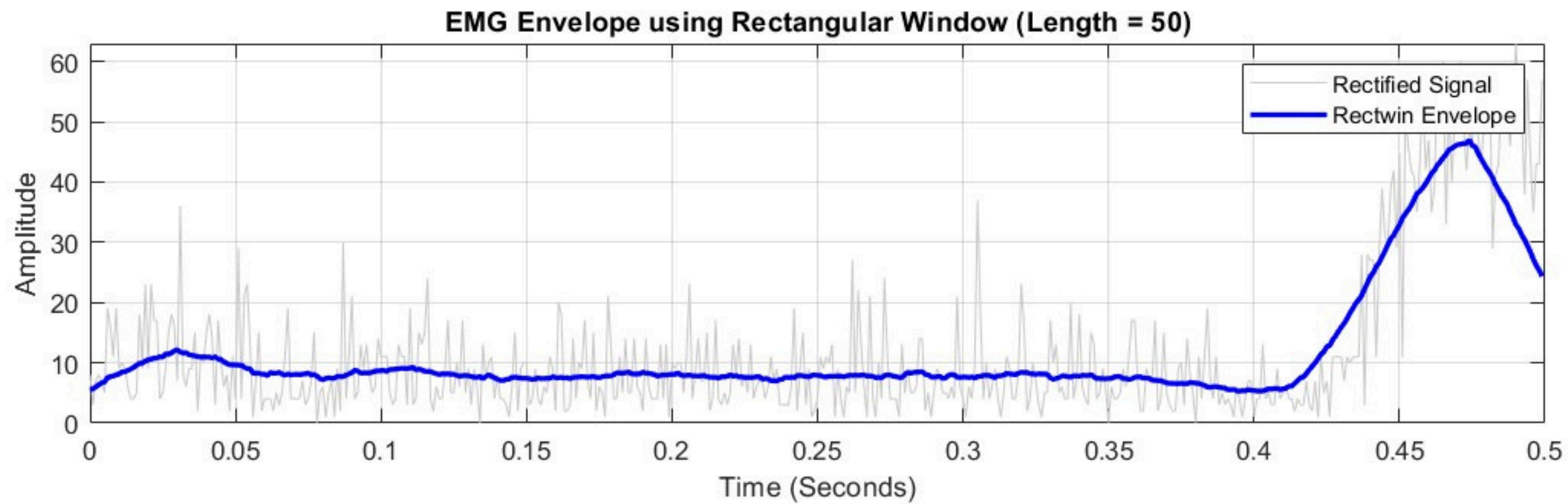
This decides the number of samples for averaging



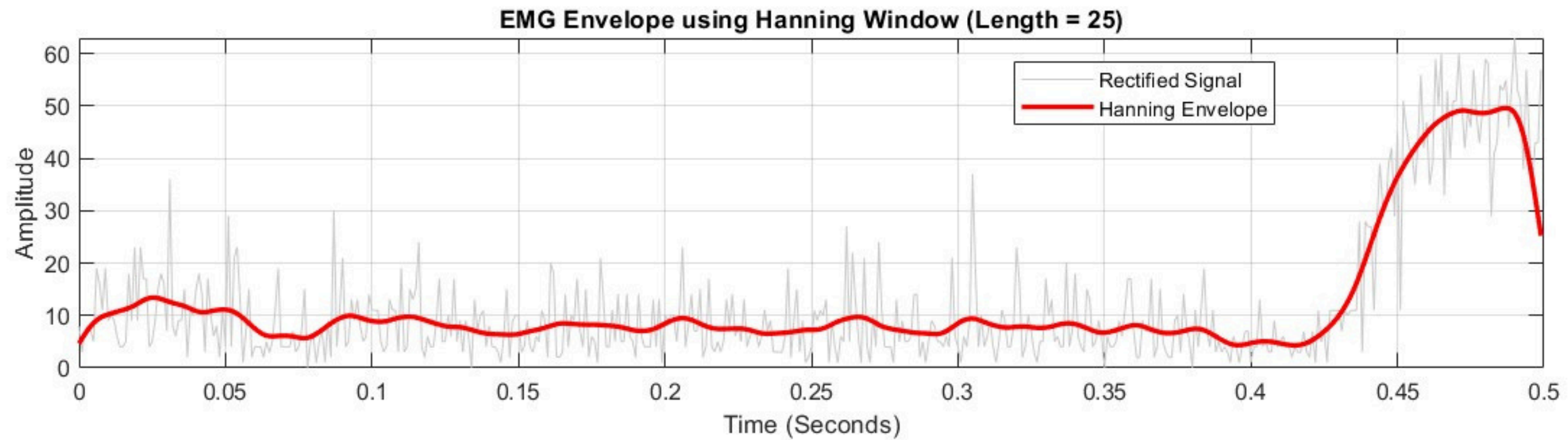
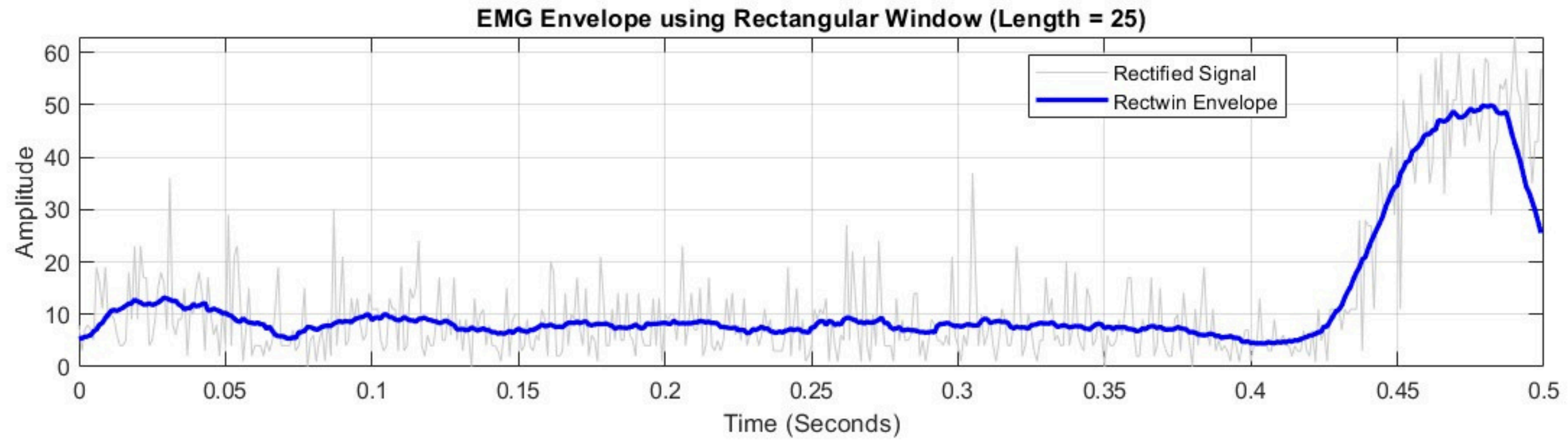
Output Timing

This decided how fast the output is generated

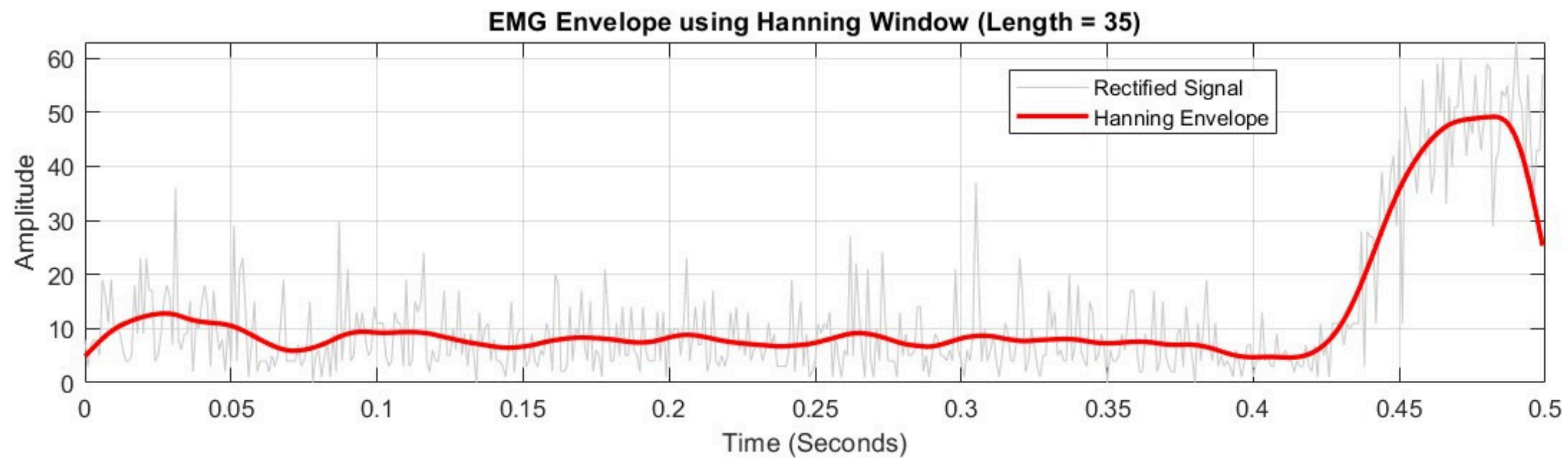
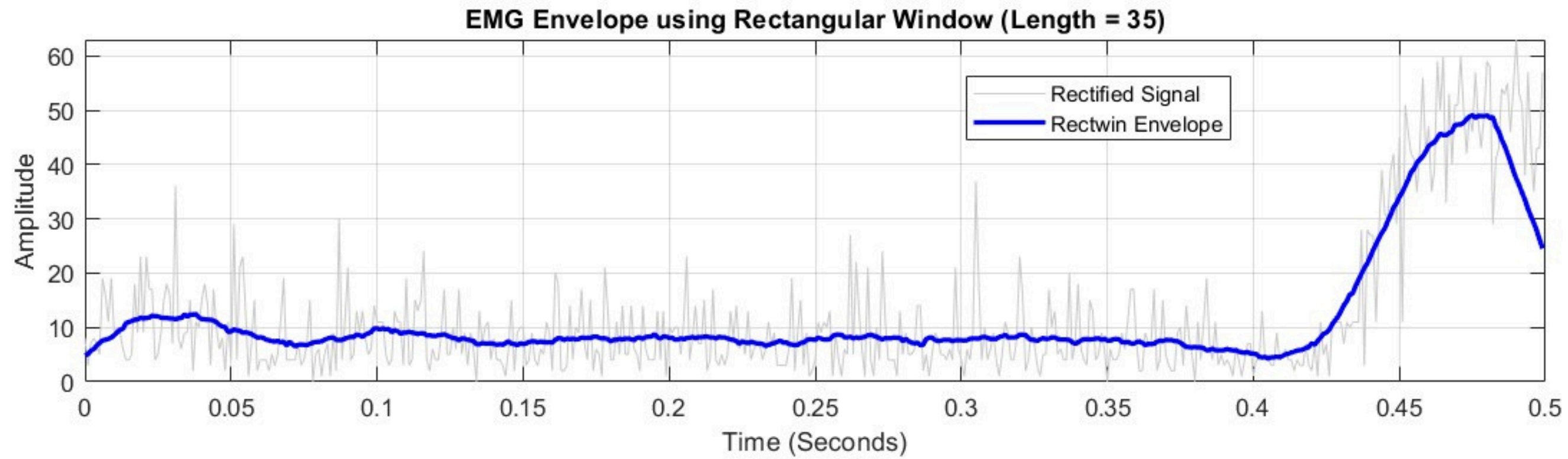
FILTERING



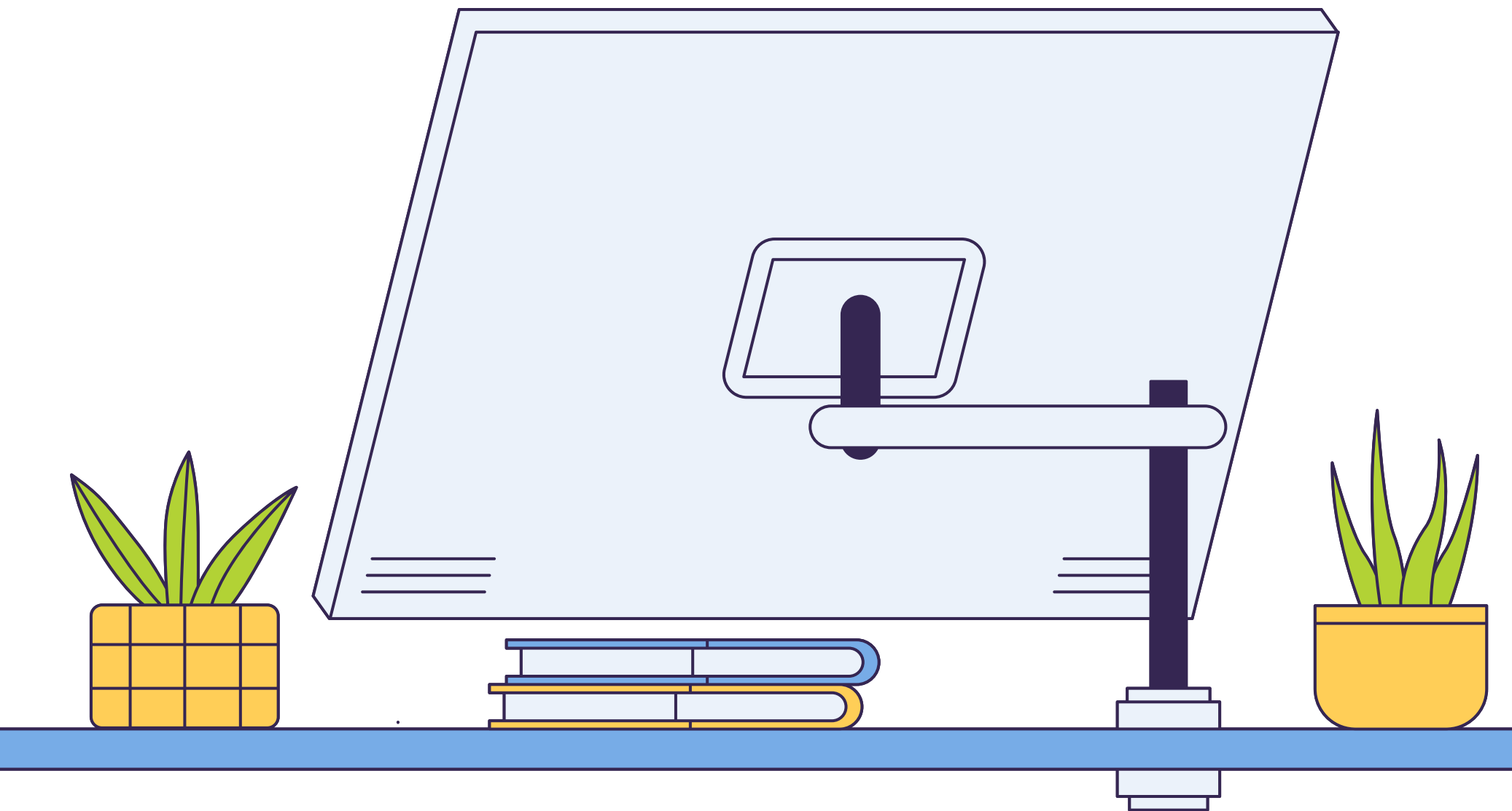
FILTERING



FILTERING



OUR VISION



Processing

We shall process the data using ML models for feature extraction and classification

Encoding

The extracted features and classification will help determine the action to be performed. This will be carried out using actuation and haptic feedback. Actuation for the actual prosthesis action

Haptic Feedback

Haptic feedback closes the loop, so it allows the brain to disengage and helps the user multitask

Thank You

